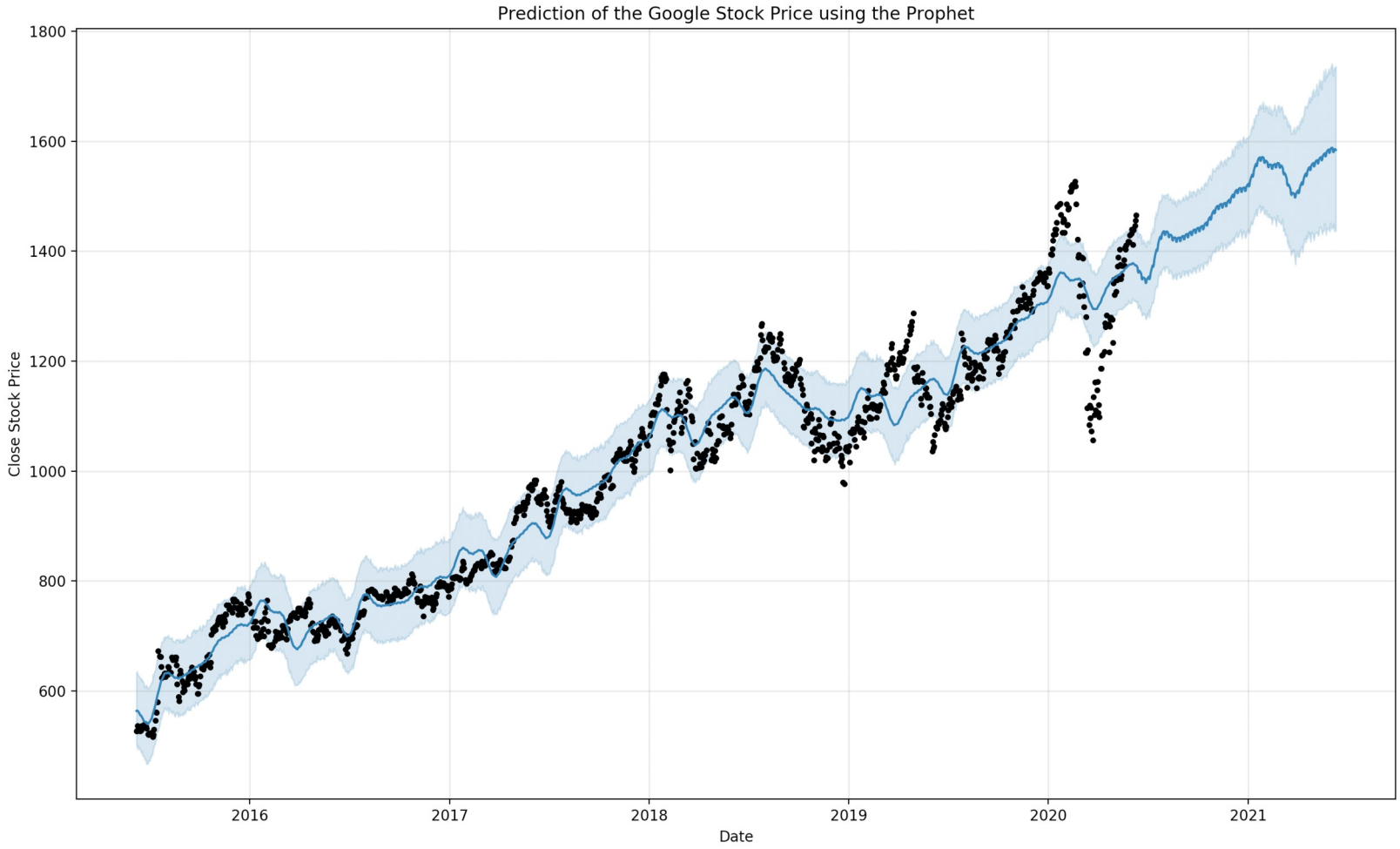
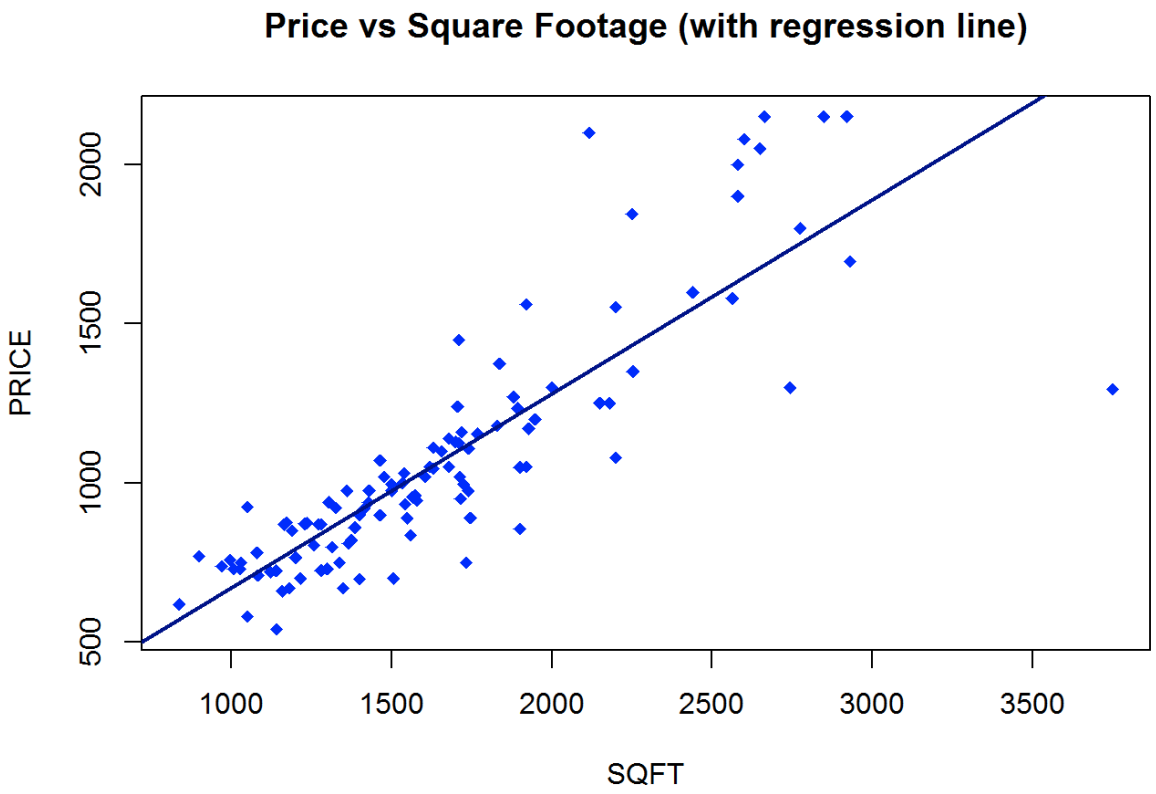
Regression


Prediction of the Google Stock Price using the Prophet
1800
1600
1400
1200
1000
800
600
Close Stock Price
2016
2017
2018
2019
2020
2021
Date

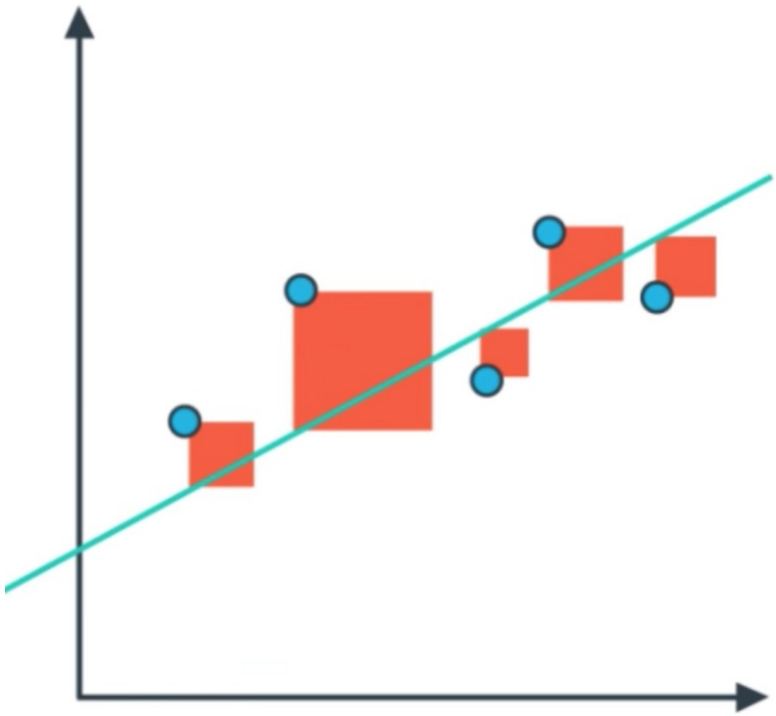

Price vs Square Footage (with regression line)
PRICE
500
1000
1500
2000
1000
1500
2000
2500
3000
3500
SQFT

	id	date	price	bedrooms	bathrooms	sqft_living	sqft_lot	floors	waterfront	view	...	grade	sqft_above	sqft_baseme
0	7129300520	20141013T000000	221900.00	3	1.00	1180	5650	1.00	0	0	...	7	1180	0
1	6414100192	20141209T000000	538000.00	3	2.25	2570	7242	2.00	0	0	...	7	2170	400
2	5631500400	20150225T000000	180000.00	2	1.00	770	10000	1.00	0	0	...	6	770	0
3	2487200875	20141209T000000	604000.00	4	3.00	1960	5000	1.00	0	0	...	7	1050	910
4	1954400510	20150218T000000	510000.00	3	2.00	1680	8080	1.00	0	0	...	8	1680	0

Evaluation

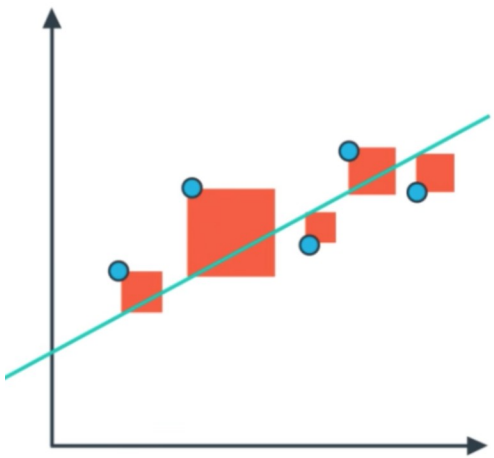
How to assess the quality of a regression model?

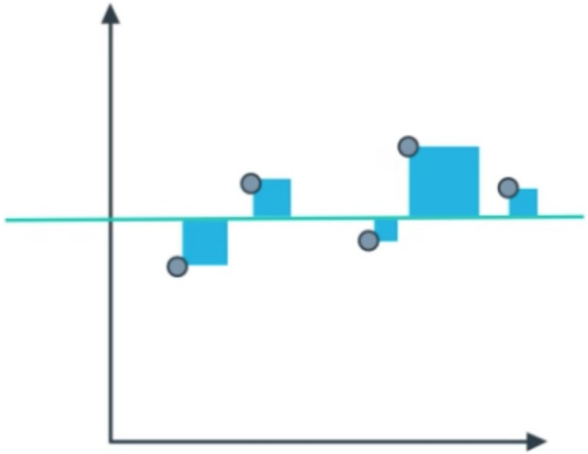
Mean square error



R2

Idea: compare your model with the simplest possible one





Y = mean (y)

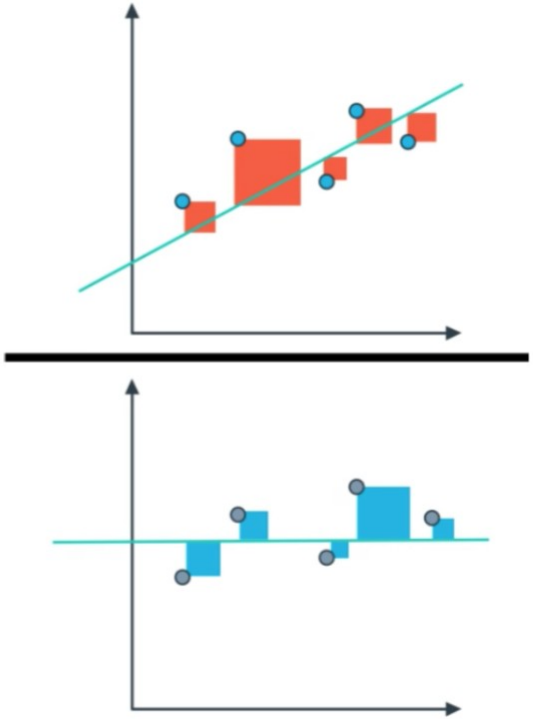
R2

Idea: compare your model with the simplest possible

Possible R2 values: from 0 to 1

Your model is:
good when R2 is close to 1
bad when R2 is close to 0

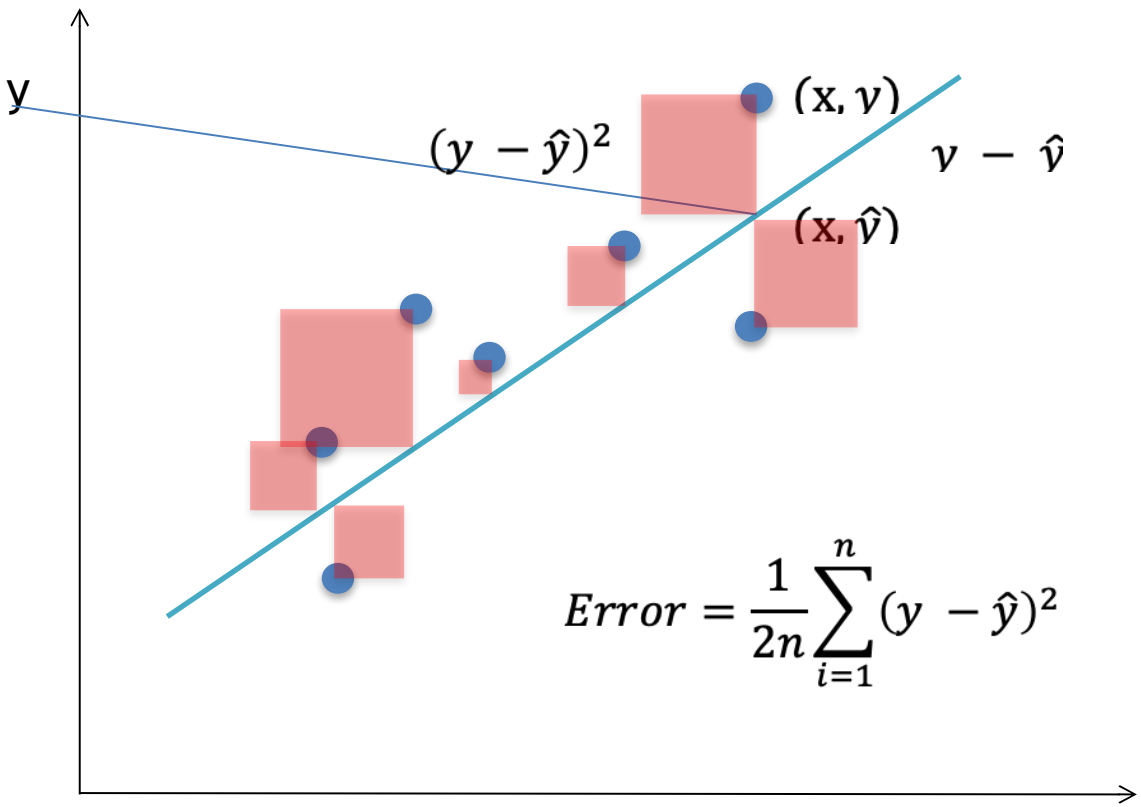
Possible values of R2: from 0 to 1

$$R2 = 1 - \frac{\text{  }}{}$$

The root of the mean square error

Root Mean Square Error

n = number of points
2 - for convenience



$$Error = \frac{1}{2n}\sum_{i=1}^{n}(y - \hat{y})^2$$